Evolutionary Control and Encoding Experiments for Mountain-Climbing Creatures

CM3020 – Artificial Intelligence (Part B Coursework)

Word Count: ~1,961 words

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Course Code: CM3020

Assessment Type: Midterm Assessment

Submission Date: 7 January 2026

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1. Introduction

This coursework evolves simulated creatures using a genetic algorithm (GA) to climb a steep central mountain in PyBullet. The focus is not only whether climbing occurs, but *why evolution fails*, *what exploits appear under weak fitness definitions*, and *how behaviour changes* as reward shaping, encoding, and simulation constraints are tightened. Fifteen experiments are conducted as a controlled sequence where each experiment targets a failure mode observed previously.

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2. Experimental Framework

The environment consists of a floor, four perimeter walls, and a central mountain (initially a smooth Gaussian pyramid). Creatures spawn at **[7, 7, 2.5]** in a corner, intentionally exposing wall/corner support exploits early. Unless changed, runs use **population = 50**, **generations = 100**, **simulation steps per evaluation = 2400**, and elitism (best preserved). Default mutation settings in standard runs are moderate (mutation rate around **0.2**, mutation amount around **0.3**) unless explicitly increased for exploration tests.

Creature evaluation tracks:

- **Final height** `last_z` (end of evaluation)
- **Maximum height** `max_z` (peak reached, logged separately)
- **Contact ratio** `contact_frames / total_frames`
- **Orientation** (quaternion → uprightness) once introduced

Per-generation logging records max/mean fitness, max height, and mean morphology size to separate genuine progress from exploits.

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3. Baseline Phase B1: What the System Optimises by Default (Experiments 1–3)

3.1 Baseline Fitness (Stable Climber)

Baseline fitness is the **Stable Climber** metric:

Fitness = Final Height × Contact Multiplier

Final height uses **`last_z`**, not peak height, so “jump and land badly” is not rewarded. Contact multiplier equals **1.0 if contact ratio > 0.8**, otherwise equals the contact ratio. “Contact” includes **floor OR mountain** contact, which stops airborne solutions but unintentionally allows wall/floor stabilisation to count as “valid stability”.

3.2 Experiment 1: Standard Baseline

Parameters changed: gene count = **3**, mutation amount = **0.3**.
Observed outcome: A "static giant" strategy emerged. Multi-link bodies lean on walls for stability, maximizing height without traversing the mountain, rewarding the metric instead of the task.

3.3 Experiment 2: Minimalist Genome

Parameters changed: gene count = **1**, mutation amount = **0.3**.
Observed outcome: A "corner leaner" appeared. Even with simple morphology, the GA wedged the creature into corners for stable height. The system correctly discovered the easiest path to reward.

3.4 Experiment 3: High Exploration

Parameters changed: gene count = **1**, mutation amount = **0.8**.
Observed outcome: Motion became chaotic. High mutation increased kinetic instability and flailing without improving progress, proving the bottleneck is lack of task pressure, not premature convergence.

B1 conclusion: the reward encourages stable wall-supported height. Exploration cannot fix missing task pressure.

Graph:

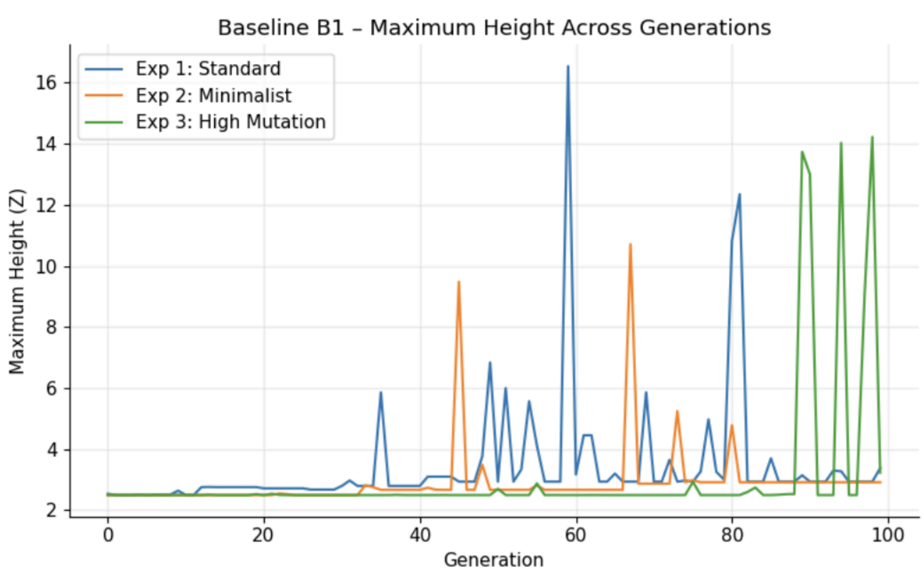

Figure 3.1: Experiments 1–3 – Max Height vs Generation

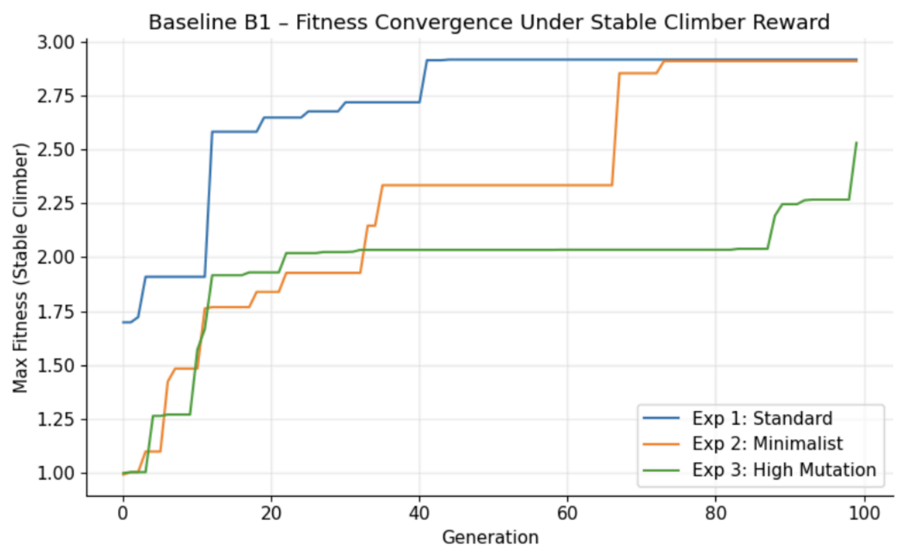

Figure 3.2: Experiments 1–3 – Fitness vs Generation

Simulation:

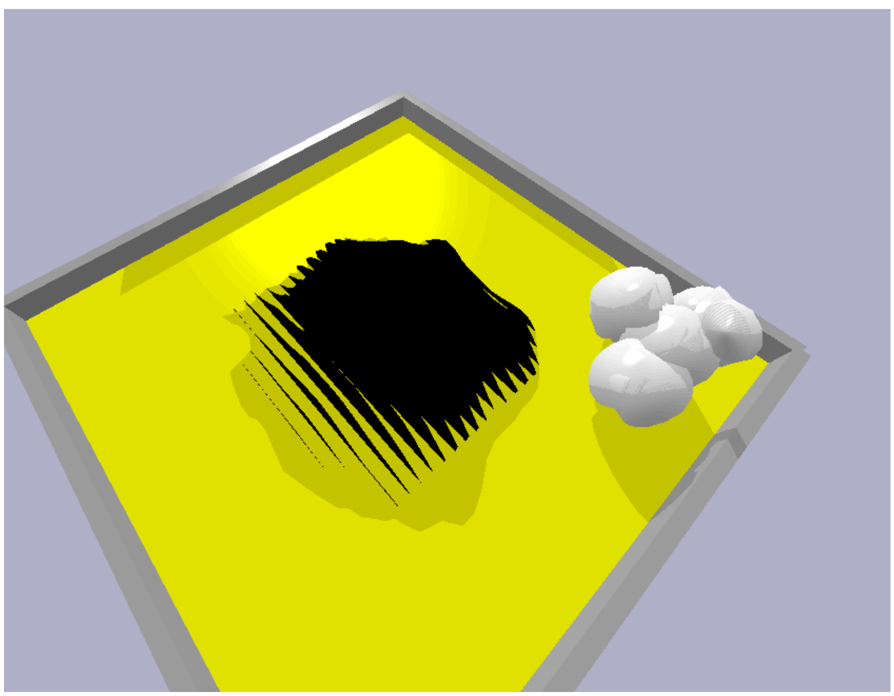

Figure 3.3: Baseline wall/corner exploitation ("static giant" or corner-supported posture)

Motivation for Experiment 4: add **directional movement pressure** and penalise size-based height cheats.

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4. Reward Shaping Phase B2–B3: Eliminating Exploits and Inducing Locomotion (Experiments 4–6)

4.1 Experiment 4: Distance Bonus + Metabolic Cost (Timber Exploit)

Fitness becomes:

Fitness = (Height × Distance Bonus × Stability) / Body Size

Distance bonus is reduction in Euclidean distance to centre (0,0), lower-bounded to avoid collapse to zero. Dividing by body size discourages large static structures.

Parameters used: gene count = **3**, mutation amount = **0.3**.
Observed outcome: wall-hugging disappears immediately, but a new exploit appears: the "timber" strategy. A tall minimal body falls forward, gains distance reward instantly, then lies prone in stable contact. The algorithm learns that "falling" is cheaper than locomotion.

Graph:

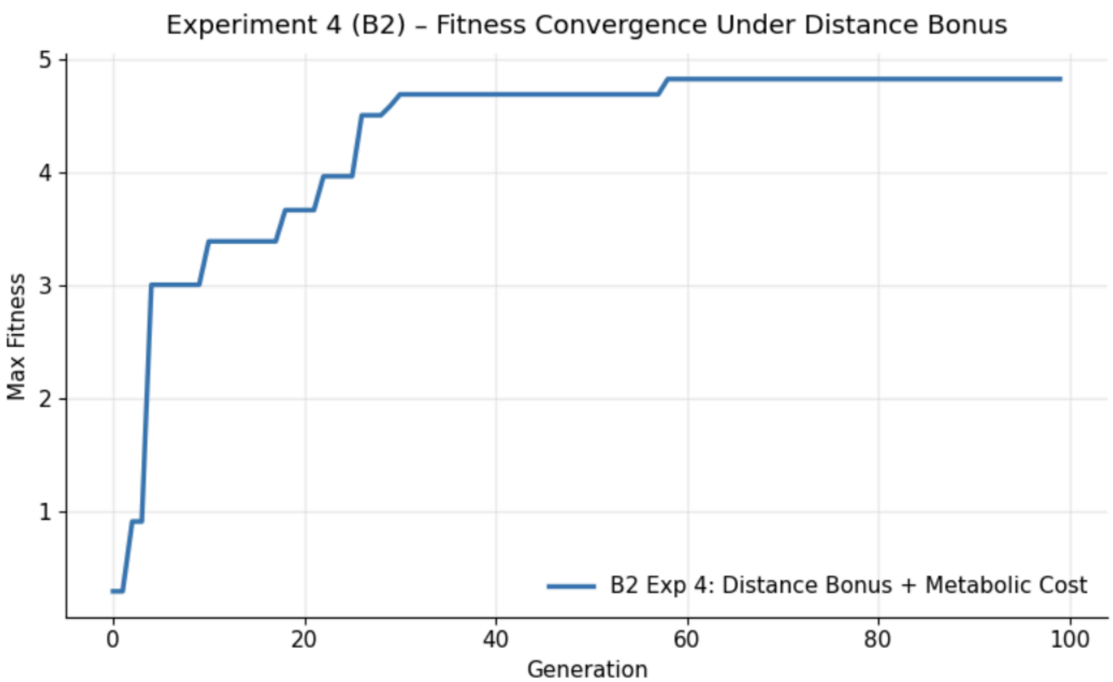

Figure 4.1: Experiment 4 – Fitness vs Generation

Simulation:

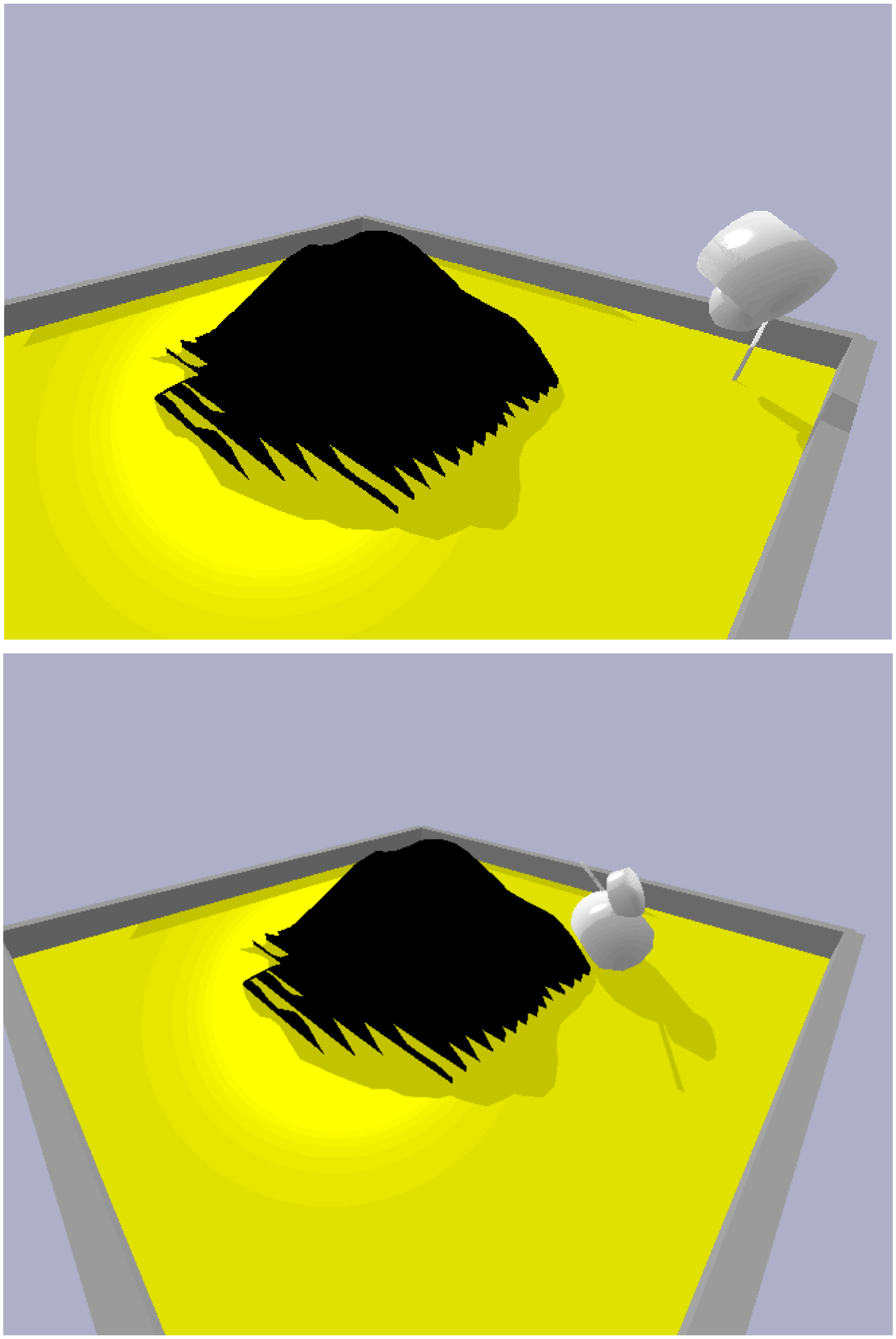

Figure 4.2: “Timber” exploit (fall-forward toward centre rewarded by distance term)

Motivation for Experiment 5: stop “lying flat” being treated as stability.

4.2 Experiment 5: Upright Orientation Penalty

Uprightness is computed from body orientation (alignment of local Z axis with world Z), clamped to [0,1] and sharpened:

upright_penalty = uprightness^4

Fitness becomes:

Fitness = (Height × Distance × Contact × Uprightness) / Body Size

Parameters: gene count = **3** (motor regime unchanged).
Observed outcome: timber exploit disappears. Creatures evolve low-profile crawling that preserves uprightness while moving toward the slope, but climbing remains limited under weak actuation.

4.3 Experiment 6: Centipede Breakthrough via Gene Count Increase

Parameter changed: gene count increased **3** → **5** (fitness unchanged).
Observed outcome: Biologically plausible locomotion emerged as multi-segment bodies produced peristaltic, centipede-like waves. Multiple contact points reduced backsliding and maintained uprightness without complex balancing. Creatures reliably reached the mountain base, marking the first “task-aligned” behavior.

Graph:

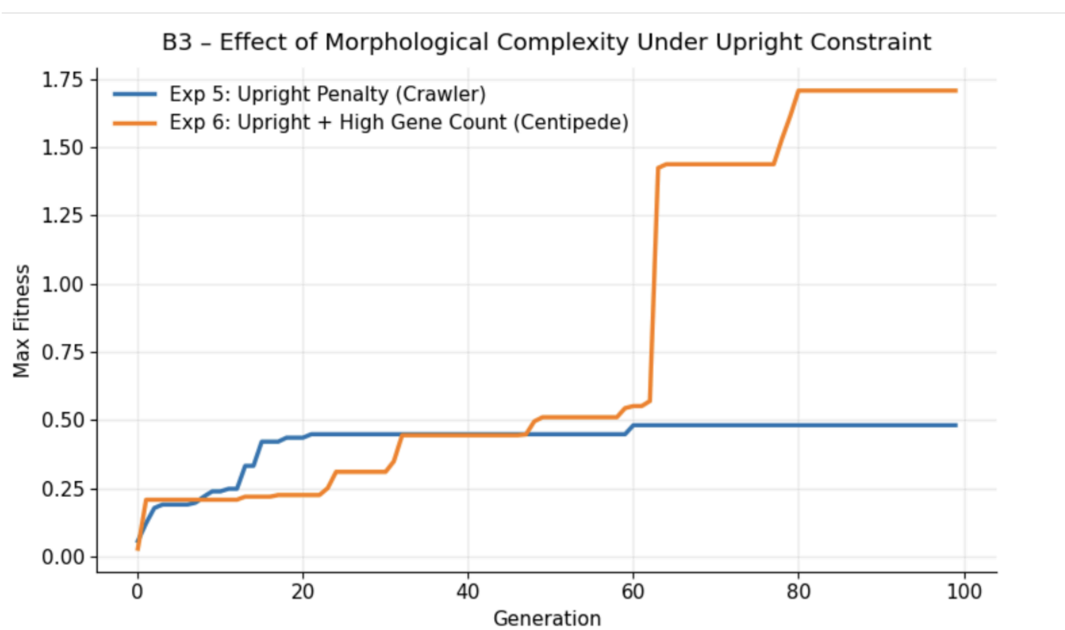

Figure 4.3: Experiments 5–6 – Fitness vs Generation

Simulation:

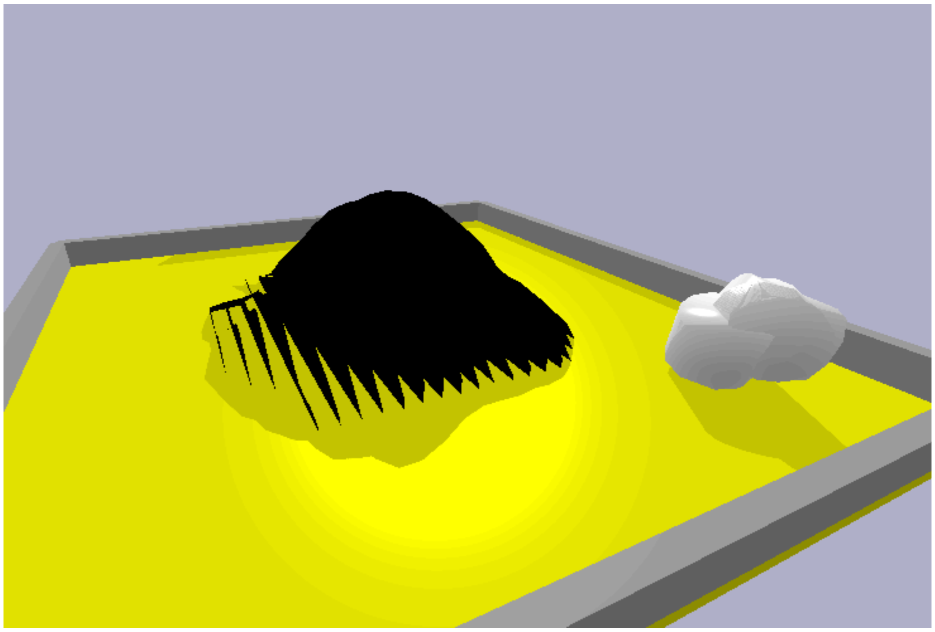

Figure 4.4: Experiment 5 – Low-profile crawling (no climb)

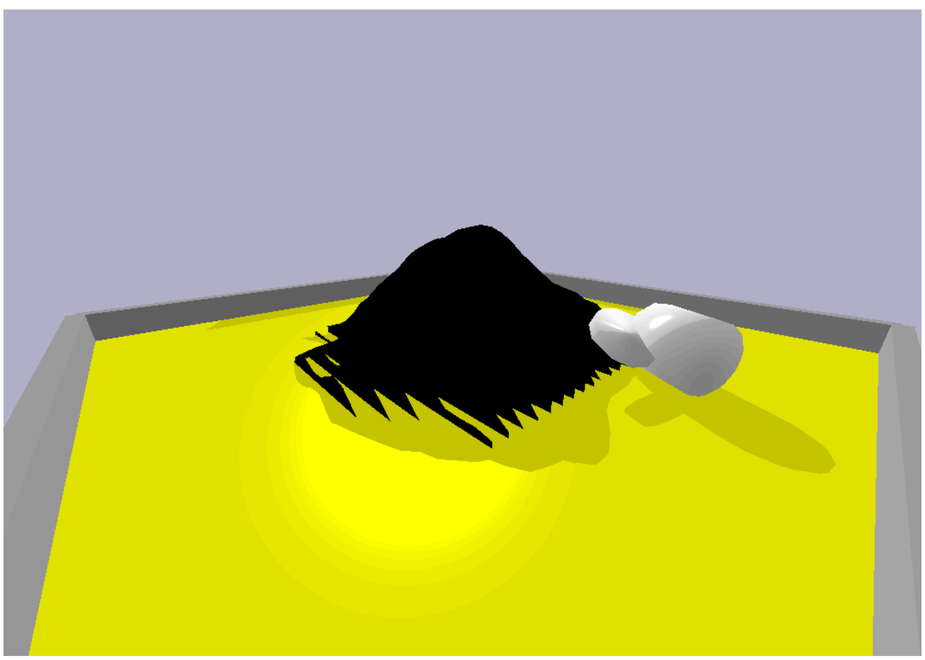

Figure 4.5: Experiment 6 – Centipede/peristaltic approach toward slope

Motivation for Experiment 7: test whether the remaining failure is search-related or a physical limit.

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5. Search Capacity Test B4: Is This an Evolution Problem? (Experiment 7)

Experiment 7: Large Search Run

Parameters changed: population **50 → 150**, generations **100 → 300** (fitness unchanged).
Observed outcome: evolution refines approach and stabilisation but does not discover sustained climbing. The run shows long periods of shrinking (metabolic cost pressure) followed by later re-expansion into longer anchors, yet behaviour remains **approach → touch slope → stall**. This suggests a performance ceiling that brute search cannot cross.

Motivation for Experiment 8: eliminate morphology exploits and simulation artefacts so the remaining failure can't be dismissed as "bad physics" or "coding noise".

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6. Encoding Scheme Redesign + Stability Engineering (Experiment 8)

Experiment 8 is a major encoding change: morphology is no longer an evolving exploit channel, so "cheating by body shape" is structurally removed.

6.1 Fixed Morphology Encoding

Morphology parameters (link length/radius/density/joint limits) are clamped to fixed constants. Joint spacing is computed analytically to ensure clean capsule-to-capsule placement (no overlaps, no gaps), removing a common source of solver explosions and artificial impulses. Shrink and grow mutations are disabled, meaning evolution acts only on control genes (waveform, amplitude, frequency, phase, and force where applicable). Gene count now directly controls the number of spine segments, not arbitrary size inflation.

6.2 Stability Engineering (Reducing Physics Noise)

To ensure the experiment measures genuine contact mechanics:

- **Passive legs** are added to each spine segment to increase contact footprint without adding new controlled DOFs (legs locked by zero-amplitude motors but physically present).
- **Collision filtering** removes destabilising internal collisions (e.g., parent-child and leg-leg interference).
- Physics tuning favours stability (e.g., smaller timestep and higher solver iterations), reducing jitter-driven artefacts and making runs repeatable.

Observed outcome: behaviour becomes consistent across runs: the creature approaches the mountain, maintains contact, and stalls due to slipping and ineffective foothold geometry. Importantly, this stall occurs without wall exploitation, without morphology bloat, and without simulation “explosions”, making it a clean baseline for physical hypothesis tests.

Graph:

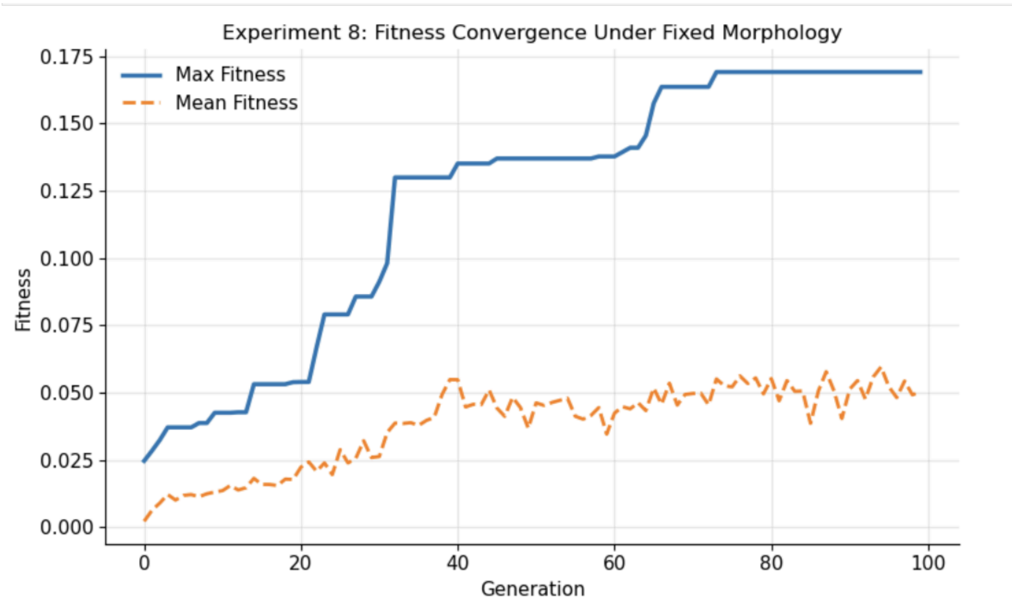

Figure 6.1: Experiment 8 – Fitness vs Generation

Simulation:

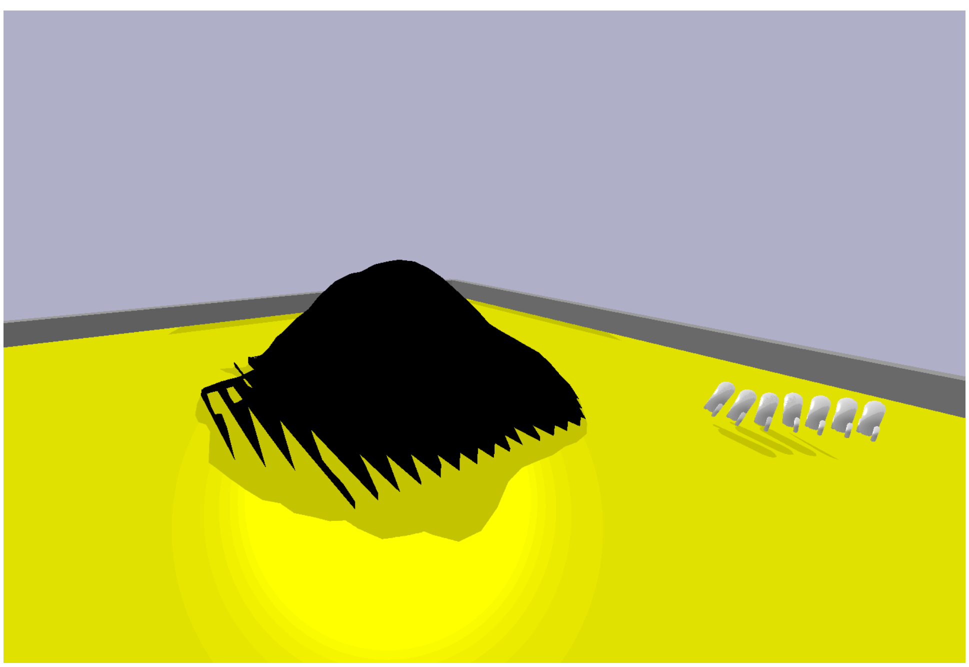

Figure 6.2: Fixed centipede geometry (clean spacing, passive stabilising legs)

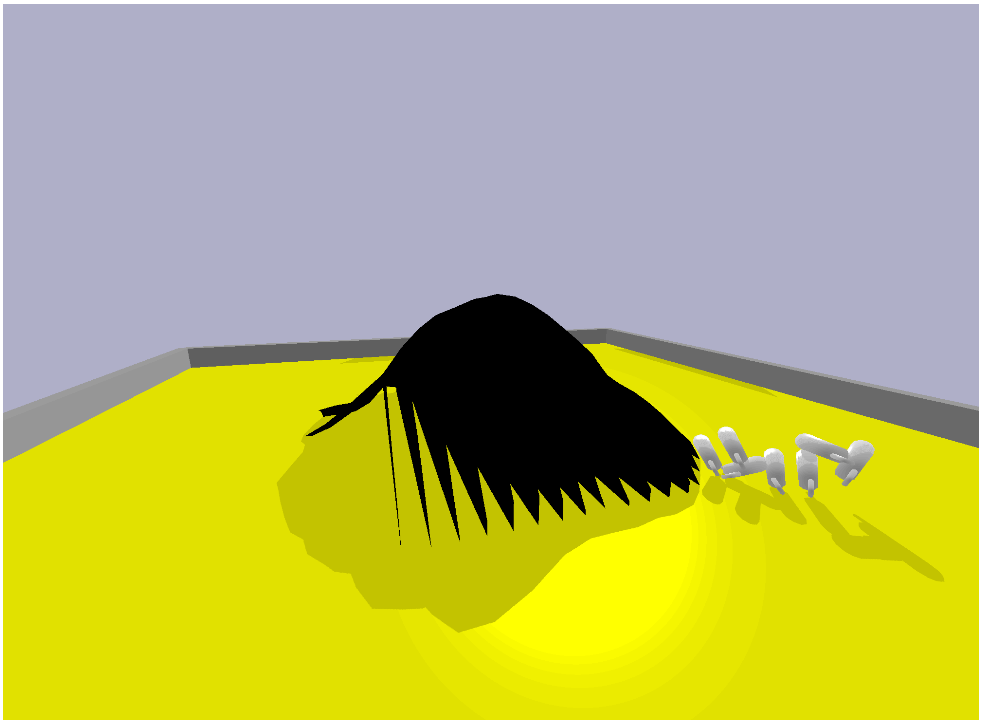

Figure 6.3: Approach → contact → stall without chaos (physics stable, no exploit)

Motivation for Experiments 9–13: with encoding stabilised, isolate whether the remaining bottleneck is friction or force.

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7. Physical Isolation + Final Validation (Experiments 9–15)

From this point onward, only one variable is changed at a time to test specific hypotheses about the remaining bottleneck, while keeping the fixed-morphology encoding and the upright/contact-based fitness stable.

7.1 Experiments 9–10: Mountain Friction Only (Grip Hypothesis)

Hypothesis: climbing fails because slope grip is too low; increasing mountain friction should enable traction.
Change applied: mountain friction increased via `changeDynamics` on the base and all links:

- lateral friction = **1.8**
- spinning friction = **0.05**
- rolling friction = **0.0**
  Floor friction unchanged to isolate slope grip alone.

Experiment 9: gene count = **7** (more segments/control DOF).
Experiment 10: gene count = **5** (simpler control).
All else unchanged.

Observed behaviour: contact quality improves, but upward progress still does not emerge. Agents repeatedly press into the slope, oscillate, and slide back into a stall. The gene-count comparison is informative: Exp 9 shows more motion patterns, but the failure mode remains the same, suggesting the missing element is not “more complicated control”, but physical leverage/contact geometry.

Table:

	Experiment	Gene Count	Mountain Friction	Max Fitness	Max Height (Z)	Outcome
0	Exp 9	7	1.8	0.106260	2.541086	Reached mountain, no sustained ascent
1	Exp 10	5	1.8	0.105059	2.505175	Reached mountain, no sustained ascent

Table 7.1: Experiments 9–10 – Friction Isolation Results

Motivation for Experiment 11: if grip is not enough, test actuation strength.

7.2 Experiment 11: Motor Force Scaling (Torque Hypothesis)

Hypothesis: failure is force-limited; stronger motors should overcome slip and generate uphill torque.
Change applied: motor force increased substantially:

- baseline force ≈ **20**
- increased force = **70**

Friction remains high to give the best chance for climbing; morphology and fitness unchanged.

Observed behaviour: Stronger actuation increased impact severity and oscillation amplitude without producing ascent. Agents failed to form stable footholds and frequently destabilized posture via collisions. Injecting more energy yielded the same outcome, proving force alone is not the missing ingredient.

Motivation for Experiments 12–13: test whether force is being wasted due to poor anchoring on the floor.

7.3 Experiments 12–13: Floor Friction Anchoring (Base-Slip Hypothesis)

Hypothesis: rear segments slip on the floor, preventing force transfer into the slope; increasing floor friction should create an anchor.
Change applied: floor lateral friction increased (mountain settings held constant):

- Experiment 12: **2.5**
- Experiment 13: **1.5**

Observed behaviour: base slip reduces and the creature looks more “grounded” during pushing. Yet the key behaviour remains: the front contacts the slope, oscillations continue, and the system settles into a stall without net ascent. This falsifies the idea that floor traction is the primary bottleneck.

Motivation for Experiments 14–15: if physical parameters don’t unlock climbing, improve control realism: brace when touching, rather than waste energy oscillating into a slide.

7.4 Experiments 14–15: Contact-Gated Control + Terrain Comparison (Final Claim Validation)

A contact-aware control architecture is introduced. Instead of blindly oscillating joints, actuation is conditioned on per-link contact state:

- Links touching the mountain switch to bracing behaviour (holding angle via **POSITION_CONTROL**).
- A small number of joints can be anchored with increased force to resist backsliding.
- This is controlled by tunable parameters (e.g., anchoring strength and number of braced joints), but the core idea is consistent: **reflex-like stabilisation on contact**.

Friction settings are explicit and reproducible:

- floor lateral friction = **0.5**
- mountain lateral friction = **1.8**

Experiment 14 terrain: gaussian_pyramid.urdf.
Experiment 15 terrain: mountain.urdf (irregular).

Observed behaviour: bracing activates correctly and reduces regression, showing the control architecture is functioning. On the Gaussian slope, movement is smoother but ascent remains minimal. On irregular terrain, contact is more frequent but produces local traps; bracing prevents sliding yet still does not generate a climbing gait. This provides a strong final validation: even when control becomes contact-aware and terrain changes, the “no sustained ascent” result persists.

Graph:

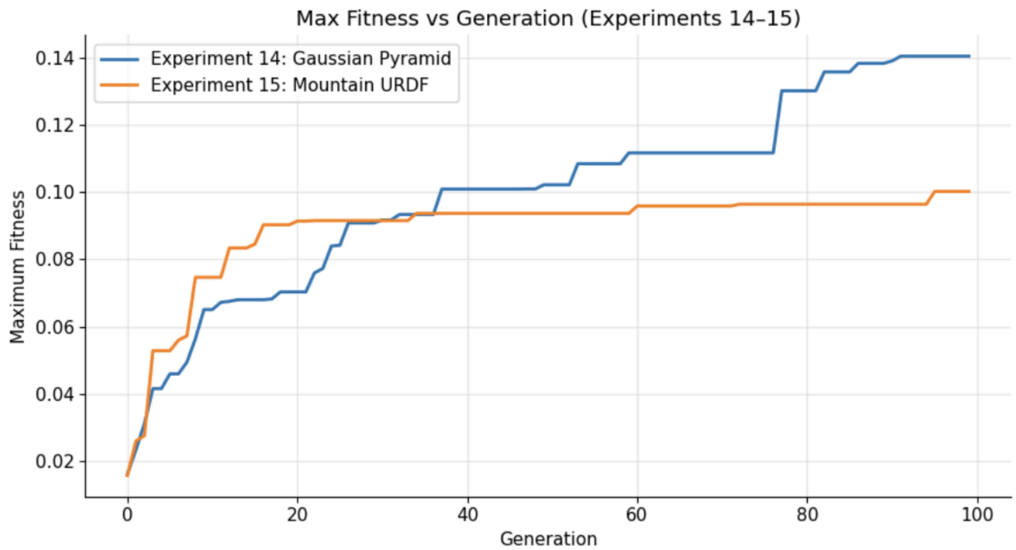

Figure 7.1: Experiments 14–15 – Fitness vs Generation

Simulation:

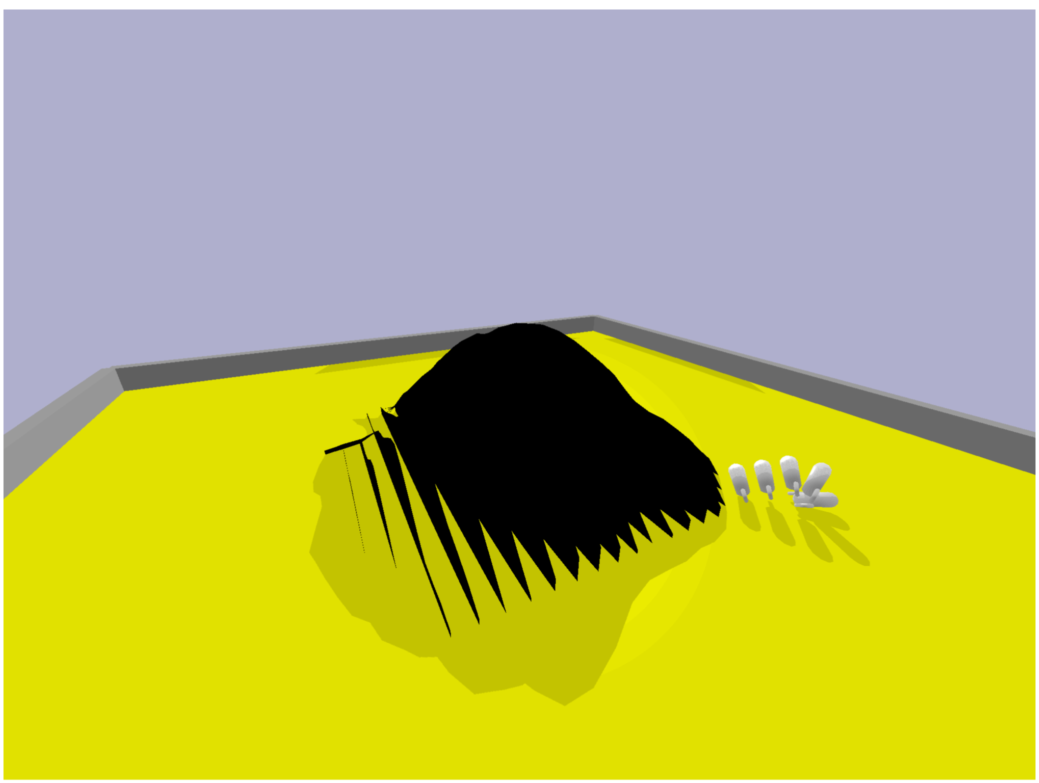

Figure 7.2: Experiment 14 – Anchoring/bracing on smooth Gaussian slope

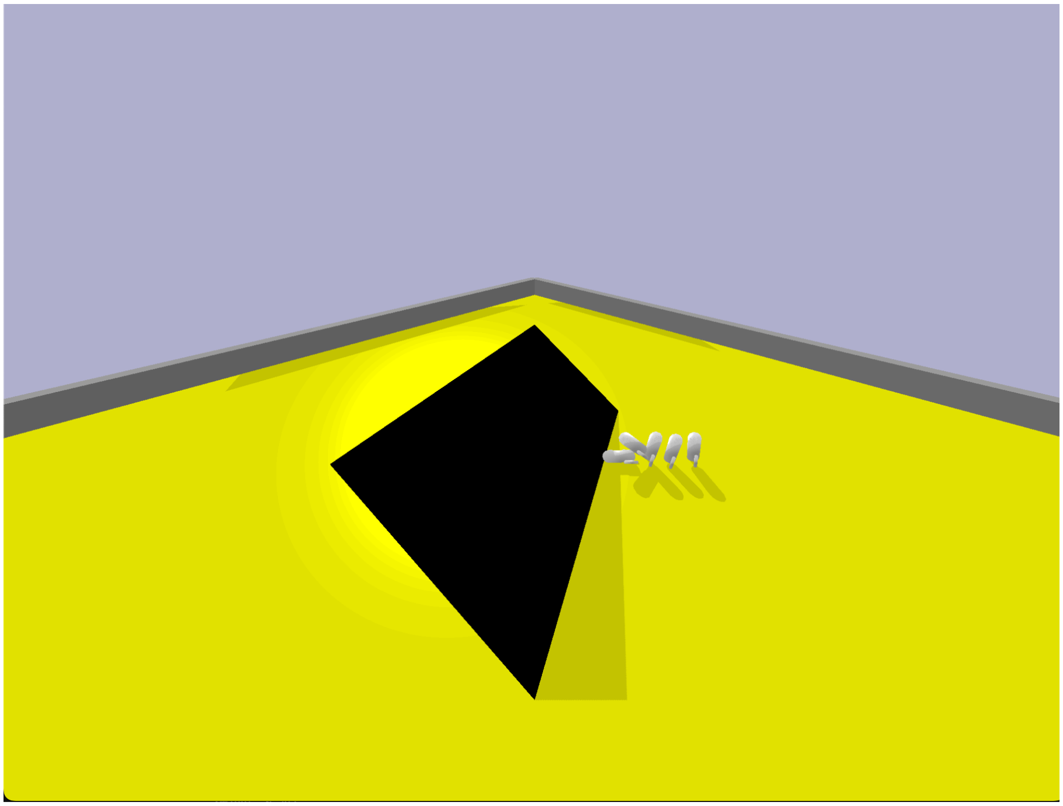

Figure 7.3: Experiment 15 – Straining/contact on irregular terrain (mountain URDF)

7.5 Final Interpretation (What 9–15 Prove)

Experiments 9–13 rule out the standard explanations under strict isolation:

- Increasing **mountain friction** improves grip but not ascent.
- Increasing **motor force** increases energy but not effective uphill traction.
- Increasing **floor friction** improves anchoring but does not change the failure mode.

Experiments 14–15 then show the same limitation even when control becomes contact-aware and terrain changes. The remaining bottleneck is therefore mechanical interaction geometry: under the fixed morphology and weak-actuation regime, the creature cannot generate stable directional footholds to convert oscillatory control into uphill progress.

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8. Overall Conclusion

This project shows that evolutionary systems in physics simulation exploit fitness weaknesses until those weaknesses are removed. Across fifteen experiments, exploit classes were identified (wall/corner support, distance-collapse “timber”, morphology anchoring) and eliminated through reward shaping, posture constraints, and a fixed-morphology encoding with stabilised simulation.

Although biologically plausible locomotion and anchoring emerged, sustained climbing did not. Controlled isolation tests confirm that increasing mountain friction, motor force, and floor friction does not change the qualitative outcome. Final contact-gated control and terrain comparison demonstrate the same limit under improved control logic.

The main result is therefore a clean, defensible claim: evolution can optimize within physical constraints, but it cannot bypass contact mechanics and actuation limits without fundamentally different interaction geometry (e.g., hooks/compliance/adhesion) or altered terrain/physics.
